Pigeonhole principle: Simple form I

Theorem

If $n+1$ objects are put into n boxes, then at least one box contains two or more of the objects.

If each of the n boxes contains at most one of the objects, then the total number of objects is at most n. Since we start with $n+1$ objects, some box contains at least two of the objects.

Application 1.

Among 13 people there are two who have their birthdays in the same month.

Application 2.

There are n married couples. How many of the $2n$ people must be selected in order to guarantee that one has selected a married couple?



Pigeonhole principle: Simple form II

Other principles related to the pigeonhole principle

- If n objects are put into n boxes and no box is empty, then each box contains exactly one object.
- If n objects are put into n boxes and no box gets more than one object, then each box has an object in it.

More abstract formulations of the three principles enunciated thus far

Let X and Y be finite sets and let $f : X \to Y$ be a function from X to Y.

- If X has more elements than Y, then f is not one to one.
- If X and Y have the same number of elements and f is onto, then f is one to one.
- If X and Y have the same number of elements and f is one to one, then f is onto.



Pigeonhole principle: Simple form III

Application 3.

Given m integers $a_1, a_2, \cdots, a_m$, there exist integers k and l with $0 \leq k < l \leq m$ such that $a_{k+1} + a_{k+2} + \ldots + a_l$ is divisible by m.

1. Consider the m sums

$$a_1, a_1 + a_2, a_1 + a_2 + a_3, \ldots, a_1 + a_2 + a_3 + \cdots + a_m.$$

2. If any of these sums is divisible by m, then the conclusion holds.
3. Thus, we may suppose that each of these sums has a nonzero remainder when divided by m, and so a remainder equal to one of $1, 2, \ldots, m-1$.
4. There are m sums and only $m-1$ remainders, two of the sums have the same remainder when divided by m.
5. Therefore, there are integers k and l with $k < l$ such that $a_1 + a_2 + \cdots + a_k$ and $a_1 + a_2 + \cdots + a_l$ have the same remainder r when divided by m,

$$a_1 + a_2 + \cdots + a_k = bm + r, \quad a_1 + a_2 + \cdots + a_l = cm + r.$$



Pigeonhole principle: Simple form IV

6. Subtracting, we find that $a_{k+1} + \cdots + a_l = (c - b)m$, thus, $a_{k+1} + \cdots + a_l$ is divisible by m.

Example

Let $m = 7$ and let our integers be 2, 4, 6, 3, 5, 5, and 6. Computing the sums as above, we get 2, 6, 12, 15, 20, 25, and 31 whose remainders when divided by 7 are, respectively, 2, 6, 5, 1, 6, 4, and 3. We have two remainders equal to 6, and this implies the conclusion that $6 + 3 + 5 = 14$ is divisible by 7.

Application 4.

A chess master who has 11 weeks to prepare for a tournament decides to play at least one game every day but, in order not to tire himself, he decides not to play more than 12 games during any calender week. Show that there exists a succession of (consecutive) days during which the chess master will have played *exactly* 21 games.



Pigeonhole principle: Simple form V

1. Let a_1 be the number of games played on the first day, etc.. The sequence of numbers $a_1, a_2, \ldots, a_{77}$ is a strictly increasing sequence since at least one game is played each day. Moreover, $a_1 \geq 1$, and since at most 12 games are played during any one week, $a_{77} \leq 12 \times 11 = 132$. Hence,

$$1 \leq a_1 < a_2 < \cdots < a_{77} \leq 132.$$

2. The sequence $a_1 + 21, a_2 + 21, \ldots, a_{77} + 21$ is also a strictly increasing sequence,

$$22 \leq a_1 + 21 < a_2 + 21 < \cdots < a_{77} + 21 \leq 132 + 21 = 153.$$

3. Thus each of the 154 numbers

$$a_1, a_2, \ldots, a_{77}, a_1 + 21, a_2 + 21, \ldots, a_{77} + 21$$

is an integer between 1 and 153.

4. It follows that two of them are equal.



Pigeonhole principle: Simple form VI

5. Since no two of the numbers $a_1, a_2, \ldots, a_{77}$ are equal and no two of the numbers $a_1 + 21, a_2 + 21, \ldots, a_{77} + 21$ are equal, there must be an i and a j such that $a_i = a_j + 21$.
6. Therefore on days $j + 1, j + 2, \ldots, i$ the chess master played a total of 21 games.

Application 5.

From the integers $1, 2, \ldots, 200$, we choose 101 integers. Show that, among the integers chosen, there are two such that one of them is divisible by the other.

By factoring out as many 2's as possible, we see that any integer can be written in the form $2^k \times a$ where $k \geq 0$ and a is odd. For an integer between 1 and 200, a is one of the 100 numbers $1, 3, 5, \ldots, 199$. Thus among the 101 integers chosen, there are two having a's of equal value when written in this form. Let these two numbers be $2^r \times a$ and $2^s \times a$. If $r < s$, then the second number is divisible by the first.



Pigeonhole principle: Simple form VII

Definition

Two positive integers m and n are said to be *relatively prime* if their greatest common divisor is 1.

Application 6. (*Chinese remainder theorem*)

Let m and n be relatively prime positive integers, and let a and b be integers where $0 \leq a \leq m-1$ and $0 \leq b \leq n-1$. Then there is a positive integer x such that the remainder when x is divided by m is a, and the remainder when x is divided by n is b; that is, x can be written in the form $x = pm + a$ and also in the form $x = qn + b$ for some integers p and q.

Consider the n integers

$$a, m+a, 2m+a, \ldots, (n-1)m+a,$$



Pigeonhole principle: Simple form VIII

each of these integers has remainder a when divided by m. Suppose that two of them had the same remainder r when divided by n. Let the two numbers be $im + a$ and $jm + a$ where $0 \leq i < j \leq n - 1$. Then there are integers q_i and q_j such that

$$im + a = q_i n + r, \qquad jm + a = q_j n + r.$$

We get

$$(j - i)m = (q_j - q_i)n.$$

n is a factor of the number $(j - i)m$. Since n has no common factor other than 1 with m, it follows that n is a factor of $j - i$. However, $0 \leq i < j \leq n - 1$ implies that $0 < j - i \leq n - 1$, and hence n cannot be a factor of $j - i$. This contradiction arises from our supposition that two of the numbers

$$a, m + a, 2m + a, \ldots, (n - 1)m + a,$$

has the same remainder when divided by n. We conclude that each of these n numbers has a different remainder when divided by n. By pigeonhole principle



Pigeonhole principle: Simple form IX

each of the n numbers $0, 1, \ldots, n-1$ occurs as a remainder, in particular, the number b does. Let p be the integer with $0 \leq p \leq n-1$ such that the number $x = pm + a$ has remainder b when divided by n. Then, for some integer q

$$x = qn + b.$$

So $x = pm + a$ and $x = qn + b$, and x has the required properties.



Pigeonhole principle: Strong form I

Theorem

Let $q_1, q_2, \ldots, q_n$ be positive integers. If

$$q_1 + q_2 + \cdots + q_n - n + 1$$

objects are put into n boxes, then either the first box contains at least q_1 objects, or the second box contains at least q_2 objects, ..., or the nth box contains at leat q_n objects.

Proof

Suppose that we distribute $q_1 + q_2 + \cdots + q_n - n + 1$ objects among n boxes. If for each $i = 1, 2, \ldots, n$ the ith box contains fewer than q_i objects, then the total number of objects in all boxes does not exceed

$$(q_1 - 1) + (q_2 - 1) + \cdots + (q_n - 1) = q_1 + q_2 + \cdots + q_n - n.$$

Since this number is on less than the number of objects distributed, we conclude that for some $i = 1, 2, \ldots, n$ the ith box contains at least q_i objects.



Pigeonhole principle: Strong form II

Special case when $q_1, q_2, \ldots, q_n$ are all equal to some integer r

- If $n(r-1)+1$ objects are put into n boxes, then at least one of the boxes contains r or more of the objects. Equivalently,
- If the average of n nonnegative integers $m_1, m_2, \cdots, m_n$ is greater than $r-1$, that is,

$$\frac{m_1 + m_2 + \cdots + m_n}{n} > r - 1,$$

then at least one of the integers is greater than or equal to r.



Pigeonhole principle: Strong form III

Another averaging principle

If the average of n nonnegative integers $m_1, m_2, \cdots, m_n$ is less than $r+1$, that is,

$$\frac{m_1 + m_2 + \cdots + m_n}{n} < r + 1,$$

then at least one of the integers is less than $r+1$.

Application 7.

A basket of fruit is being arranged out of apples, bananas, and oranges. What is the smallest number of pieces of fruit that should be put in the basket in order to guarantee that either there are at least 8 apples or at least 6 bananas or at least 9 oranges?

Solution

$8 + 6 + 9 - 3 + 1 = 21$.



Pigeonhole principle: Strong form IV

Yet another averaging principle

If the average of n nonnegative integers $m_1, m_2, \cdots, m_n$ is at least equal to r, then at least one of the integers $m_1, m_2, \cdots, m_n$ satisfies $m_i \geq r$.

Application 8.

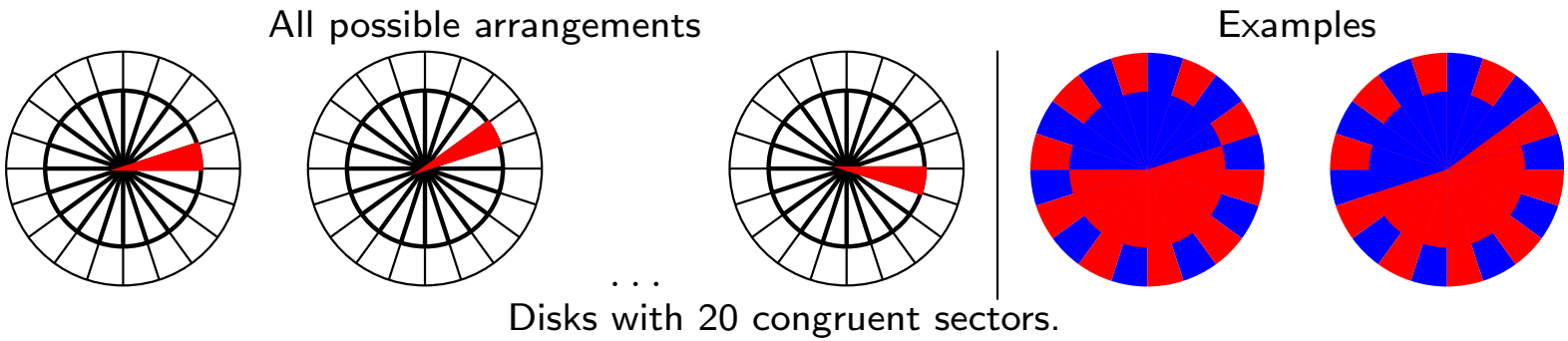
Two disks, one smaller than the other, are each divided into 200 congruent sectors. In the larger disk, 100 of the sectors are chosen arbitrarily and painted red; the other 100 sectors are painted blue. In the smaller disk, each sector is painted either red or blue with no stipulation on the number of red and blue sectors. The small disk is then placed on the larger disk so that their centers coincide. Show that it is possible to align the two disks so that the number of sectors of the small disk whose color matches the corresponding sector of the larger disk is at least 100.



Pigeonhole principle: Strong form V

Solution

Since the large disk has 100 sectors of each of the two colors, each sector of the small disk will match in color the corresponding sector of the large disk in exactly 100 of the 200 possible positions. Thus the total number of color matches over all the positions equals the number of sectors of the small disk multiplied by 100, and this equals 20,000. Therefore, the average number of color matches per position is $20,000/200 = 100$. So there must be some position with at least 100 color matches.



Pigeonhole principle: Strong form VI

Application first discovered by Erdös and Szekeres.

Application 9.

Show that every sequence $a_1, a_2, \ldots, a_{n^2+1}$ of n^2+1 real numbers contains either an increasing subsequence of length $n+1$ or a decreasing subsequence of length $n+1$.

1. For each $k = 1, 2, \ldots, n^2+1$ let m_k be the length of the longest increasing subsequences which begins with a_k.
2. If any of $m_1, m_2, \ldots, m_{n^2+1}$ is greater than or equal to $n+1$, then the conclusion holds.
3. Or the numbers $m_1, m_2, \ldots, m_{n^2+1}$ are n^2+1 integers between 1 and n.



Pigeonhole principle: Strong form VII

4. By the strong form of the pigeonhole principle, $n+1$ of the numbers are equal. Let

$$m_{k_1} = m_{k_2} = \cdots = m_{k_{n+1}} \quad (k_1 < k_2 < \cdots < k_{n+1}),$$

where $1 \leq k_1 < k_2 < \cdots < k_{n+1} < n^2 + 1$.

5. Suppose that for some $i = 1, 2, \ldots, n$, $a_{k_i} < a_{k_{i+1}}$. Then we could have a longest increasing subsequence beginning with $a_{k_{i+1}}$ and put a_{k_i} in front to obtain an increasing subsequence beginning with a_{k_i}. This implies that $m_{k_i} > m_{k_{i+1}}$, we conclude $a_{k_i} \geq a_{k_{i+1}}$. This is true for each $i = 1, 2, \ldots, n$, We have

$$a_{k_1} \geq a_{k_2} \geq \cdots \geq a_{k_{n+1}}.$$

6. We conclude that $a_{k_1}, a_{k_2}, \ldots, a_{k_{n+1}}$ is a decreasing subsequence of length $n+1$.



Pigeonhole principle: Strong form VIII

An amusing formulation of Application 9 is the following:

> *Suppose that n^2+1 people are lined up shoulder to shoulder in a straight line. Then it is possible to choose $n+1$ of the people to take one step forward so that going from left to right their heights are increasing (or decreasing).*



A theorem of Ramsey I

The most popular and easily understood instance of Ramsey's theorem:

> *Of six (or more) people, either there are three, each pair of whom are acquainted, or there are three, each pair of whom are unacquainted.*

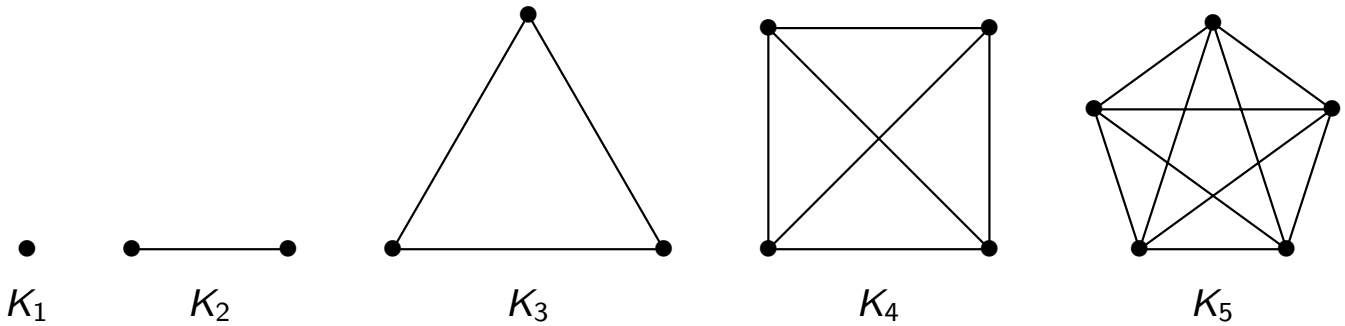
Abstractly description

$$K_6 \to K_3, K_3. \tag{2.1}$$

In general, we mean by K_n a set of n objects and all the pairs of these objects. We often refer to K_3 as a *triangle*.

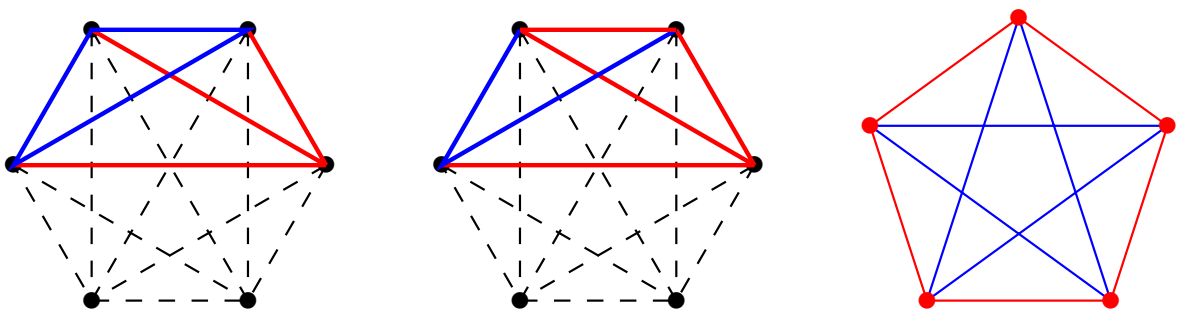


A theorem of Ramsey II



Illustrations for K_n $(n = 1, 2, 3, 4, 5)$.

$K_6 \to K_3, K_3$ is the assertion that no matter how the edges of K_6 are colored with the colors red and blue, there is always a red K_3 or a blue K_3, in short a monochromatic triangle.



Assertion $K_6 \to K_3, K_3$ is true while $K_5 \to K_3, K_3$ is false.

A theorem of Ramsey III

More generally, Ramsey's theorem, still not in its full generality, asserts

if $m \geq 2$ and $n \geq 2$ are integers, then there is a positive integer p such that

$$K_p \to K_m, K_n$$

The *Ramsey's number* $r(m, n)$ is the smallest integer p such that $K_p \to K_m, K_n$.

$r(3,3) = 6$,
$r(3,4) = r(4,3) = 9$,
$r(3,5) = r(5,3) = 14$,
$r(3,6) = r(6,3) = 18$,
$r(3,7) = r(7,3) = 23$,
$r(3,8) = r(8,3) = 28$,
$r(3,9) = r(9,3) = 36$,
$40 \leq r(3,10) = r(10,3) \leq 43$,
$r(4,4) = 18$,
$r(4,5) = r(5,4) = 25$,
$43 \leq r(5,5) \leq 49$.

Known facts about nontrivial Ramsey numbers $r(m, n)$. See also Ma et al. (2026).



A theorem of Ramsey IV

$r(2,n) = n$, $r(m,2) = m$. These are the *trivial Ramsey numbers*.

$(r(2,n) \leq n)$: If we color the edges of K_n either red or blue, then either some edge is colored red (a red K_2), or all edges are blue (a blue K_n).

$(r(2,n) > n-1)$: If we color all the edges of K_{n-1} blue, then we have neither a red K_2 nor a blue K_n.

The Ramsey number $r(n_1, n_2, \cdots, n_k)$ are defined in a similar way, and Ramsey's theorem in its full generality for pairs asserts that these numbers exist; that is, there is an integer p such that

$$K_p \to K_{n_1}, K_{n_2}, \cdots, K_{n_k}.$$

There is a more general form of Ramsey's theorem in which pairs (subsets of two elements) are replaced by subsets of t elements for some fixed integers $t \geq 1$. Let K_n^t denote the collection of all subsets of t elements of a set of n elements.



A theorem of Ramsey V

Theorem

The general form of Ramsey's theorem – Given integers $t \geq 1$ and integers $q_1, q_2, \ldots, q_k \geq t$, there exists an integer p such that

$$K_p^t \to K_{q_1}^t, K_{q_2}^t, \ldots, K_{q_k}^t.$$

The smallest such integer p is the *Ramsey number*

$$r_t(q_1, q_2, \ldots, q_k).$$

Suppose $t = 1$. Thus, by the strong form of the pigeonhole principle,

$$r_1(q_1, q_2, \ldots, q_k) = q_1 + q_2 + \ldots + q_k - k + 1.$$

This demonstrates that Ramsey's theorem is a generalization of the strong form of the pigeonhole principle.

The determination of the general Ramsey numbers is a difficult problem. However, it is not difficult to see that $r_t(t, q_2, \ldots, q_k) = r_t(q_2, \ldots, q_k)$.